

Title: Encrypted, anonymized, and obfuscated communication of status and state changes for electronic physical product Identifiers.

Abstract: A process, In a system or network of computers where two or more parties wish to isolate and protect their information about electronic physical product identifiers but wish to share the state, status, part or all of their records for that item if another correctly credentialed participant presents an obscured but identical set of identifiers to those of an item of which they have knowledge.

Background: The 2014 Drug Supply chain Security Act (the DSCSA) Requires that all Pharmaceutical product be unique labeled and that correctly credentials third parties can verify those labels with the manufacturer during the entirety of a product's lifecycle. As the DSCSA does not stipulate the use of a centralized database, each manufacturer is responsible for verifying their own product. Third-party IT suppliers can be used to manage these verifications for manufacturers. The DSCSA requires that by 2023 all product be fully traceable, from its manufacture to dispensing, as participants do not wish to reveal their full inventory activity to each other the DSCSA allows for each trading party to release information about a product to the next trading party and to vouch for its origins. This leads to two issues, how best to communicate the good state of a product from one party to the next and how to backtrace such status over time and across all parties if a product is found to be tainted or have been tampered with. This invention addresses the problem as described, and allows for the state, current price, location or any other useful metadata to be associated with an item and to be searchable by only authorized parties while still managing to obscure any other specific or aggregated data about the product or each filing parties activities.

Summary of the Disclosure: In an electronic system and method, string^a is a physical product identifier; a sequence of identifying data that can distinguish one item from all other alike items in a group or batch can identify that group or batch, from any other group or batch, and can identify this product from any other physically alike product or version of the product from the same manufacturer and can further distinguish this manufacturer's alike product from any other manufacturers alike product.

In this system or network of computers participating members of the community record all their physical product identifiers in the same format as String^a in a private repository that also stores associated useful metadata about the products historical activity such as it's state, location, ownership, temperature or other products it has been associated with for the purpose of packing or shipping.

A static or rolling encryption Key^a known to all authorized parties within the community is combined with agreed parts of the identification string and used to calculate a unique, fixed-

length output hash^a by application of a hashing algorithm such as a SHA256 or some other mathematical method and added to each Strings records.

When a community member wishes to make an inquiry of another party about a specific item they utilize the output hash. If any other party has seen the item, they will have generated the same key output Hash and by comparison to their own records be able to understand what item the inquiry is about. If a community member has not seen that particular item, the hash will be meaningless and no details of the underlying product will have been shared.

The addition of the extra string or key has ensured that unauthorized parties who are not privy to Key^a but happen across String^a, cannot confirm its existence against a community member's records. This allows the system to exist across a series of published, replicated ledgers which could be accessible to all, such as in a blockchain or distributed ledger system, or in a similar private access blockchain or distributed ledger system where, even though parties are invited to join and their access controlled, they are still not fully trusted.

When a request for information is sent using Hash^a and presented to one or more other community members, the members return a negative response if they do not have that Hash in their records or a positive response if they do. If required, further keys can be distributed and the output hashes generated and stored. These further Keys^{b to n} can require a positive responder to include agreed additional metadata giving the system flexibility or allowing sub-groupings of authorized parties

Also, for the purpose of the system and search efficiency Hash^a could be stored within a group where the group represents a specific attribute such as, but not limited to time, location, state, ownership, or financial lien. The contents of this group may be obscured by encrypted containerization where the container only responds to a single inquiry with a single response. As the data size of the container may be ascertainable, salt can be added in the form of nonpertinent data that equalizes the data storage size of all containers, thus preventing inference of the number of records contained based on the size of the container once it resides on the blockchain or distributed ledger system.

While similar to a zero-knowledge proof this claim is unique in that as the physical product identifier is previously known to both parties in any transaction the hash Output can be generated asynchronously and stored in advance, greatly speeding up any later verification process, there is no requirement to run a series of reiterative challenges to the revealing party a single ask and answer message set will suffice to attain confirmation, and as the information in the request is held in a standard length single string the message load on the system is considerably diminished compared to sending the underlying product identity information.

Brief description of the drawings:

Figure 1A describes the process of using the data within a physical product identification string combining it with a key and calculating a non-reversible output string that is understandable to any other party who has access to both the key and the physical identification string but is meaningless to any other party.

Figure 2A describes the process of using the data within a physical product identification string combining it with a key and calculating the same non-reversible output string in order to prove to another party that you are in possession of both the Key and the product identification string.

Figure 3A describes how a recipient of a request compares the inbound request hash with their stored set of hashes and returns either a negative rejection or a positive conformation with any pre-agreed metadata.

Claims

What is claimed for:

1. An anticounterfeiting method, that allows requesting parties within a network of computers to check electronically that a label was issued by the manufacturer
 - a. By routing messages to the publicly available and accessible website URL or modified website URL of a product manufacturer and receiving a response back directly from the manufacturer, thus giving surety that the responding party is the genuine product manufacturer because the requestor has now checked that they have control over the manufacturers public URL.
 - b. By routing messages to the publicly available and accessible website URL or modified website URL of a product manufacturer and receiving a response back from the manufacturers designated third party verification systems vendor who has been forwarded the message from the manufacturer, thus giving surety that the responding party is the genuine product manufacturer because the requestor has now checked that they have control over the manufacturers public URL.

ENDS